

eRTMAC-NWIS | Pre-Spud Hazard Briefing

Duliajan-Target-01 (Planned) | Duliajan | 27.3615N 95.3290E | Planned TD 4800 m | Offset radius 10 km | Generated 2026-09-29 15:15

1. Executive summary

14 offset wells within 10 km report 48 drilling events totalling 575 h NPT. The XGBoost risk model flags 3 x 100 m intervals as HIGH risk, concentrated in Girujan Clay, Kopili Shale, Tipam Sandstone.

2. Offset well similarity ranking

#	Well	Field	Dist km	Litho	MW	Traj	Score	Events	NPT h
1	Duliajan-01	Duliajan	0.72	0.92	0.78	0.981	0.8782	4	50.3
2	Duliajan-04	Duliajan	1.17	0.929	0.82	0.989	0.8695	3	31.3
3	Duliajan-12	Duliajan	1.07	0.93	0.84	0.863	0.8605	4	39.4
4	Duliajan-07	Duliajan	1.74	0.944	0.887	0.854	0.8445	4	61.9
5	Duliajan-05	Duliajan	1.81	0.934	0.753	0.988	0.8254	5	57.4
6	Duliajan-06	Duliajan	2.99	0.928	0.98	0.891	0.8221	3	36.8
7	Duliajan-08	Duliajan	2.14	0.911	0.873	0.878	0.8191	2	39.3
8	Duliajan-10	Duliajan	2.21	0.956	0.82	0.885	0.8173	3	39.1
9	Duliajan-02	Duliajan	2.94	0.941	0.767	1.0	0.7906	5	54.9
10	Duliajan-09	Duliajan	1.28	0.959	0.513	0.833	0.7732	3	30.6
11	Duliajan-03	Duliajan	3.93	0.908	0.807	0.873	0.7417	2	31.7
12	Duliajan-11	Duliajan	2.44	0.947	0.56	0.777	0.7248	4	40.1

3. Depth hazard matrix (offset events by formation)

Formation	Top m	Loss	Kick	Stuck pipe	Caving
Namsang	450	2 / 6h	-	-	-
Girujan Clay	1150	-	-	8 / 85h	3 / 19h
Tipam Sandstone	2820	12 / 186h	1 / 9h	-	-
Barail Coal	3480	-	-	12 / 192h	3 / 20h
Kopili Shale	4150	-	3 / 30h	-	4 / 28h

4. Predicted risk by depth section (100 m)

Interval m	Formation	Risk	Explanation (SHAP)	Level
1100-1200	Girujan Clay	79%	Driven by: 0 m below formation top + stuck pipe in offset well Duliajan-02 at	HIGH
2800-2900	Tipam Sandstone	96%	Driven by: 30 m below formation top + loss in offset well Duliajan-01 at 2926	HIGH
3500-3600	Barail Coal	46%	Driven by: stuck pipe in offset well Duliajan-07 at 3463 m (22.1 h NPT) + 70 m	MODERATE
4100-4200	Kopili Shale	83%	Driven by: 0 m below formation top + kick in offset well Duliajan-12 at 4167 m	HIGH
4200-4300	Kopili Shale	58%	Driven by: kick in offset well Duliajan-12 at 4167 m (14.7 h NPT) + narrow 3.3	MODERATE

5. Recommended casing seats

Depth m	Casing	Rationale
450	13 3/8 in surface	Isolate alluvium / shallow aquifers
2790	9 5/8 in intermediate	Set above Tipam Sandstone (peak risk 96%) to protect weaker shallow shoe
3450	7 in production liner	Set above Barail Coal (peak risk 46%) to protect weaker shallow shoe
4120	5 in liner	Set above Kopili Shale (peak risk 83%) to protect weaker shallow shoe

6. Lessons learned - recommended actions

Girujan Clay (top 1150 m) - stuck pipe: 8/14 wells, avg NPT 10.6 h

[WORKED x3] Switched to KCl-polymer mud (7% KCl) to inhibit clay swelling

[WORKED x3] Jarred down with 60 klbs overpull limit, spotted pipe-lax pill
[WORKED x2] Backreamed each stand and maintained 120 rpm rotation
[FAILED x3] Pulling above 120 klbs overpull parted the string
[FAILED x1] Water-based mud without inhibitor caused bit balling

Tipam Sandstone (top 2820 m) - loss: 12/14 wells, avg NPT 15.5 h

[WORKED x4] LCM pill (4 ppb) + reduced ROP to 3 m/hr
[WORKED x4] Lowered ECD by reducing pump rate; bridged with graded LCM
[WORKED x3] Pumped 40 bbl CaCO₃ fine/medium LCM pill, reduced flow rate by 15%
[FAILED x3] Increasing mud density above 11.8 ppg worsened formation fracture
[FAILED x3] Coarse nut-plug LCM plugged MWD tool, forced trip
[FAILED x2] Continued drilling blind without LCM led to total loss

Barail Coal (top 3480 m) - stuck pipe: 12/14 wells, avg NPT 16.0 h

[WORKED x5] Backreamed each stand and maintained 120 rpm rotation
[WORKED x4] Switched to KCl-polymer mud (7% KCl) to inhibit clay swelling
[WORKED x3] Jarred down with 60 klbs overpull limit, spotted pipe-lax pill
[FAILED x6] Water-based mud without inhibitor caused bit balling
[FAILED x3] Pulling above 120 klbs overpull parted the string

Girujan Clay (top 1150 m) - caving: 3/14 wells, avg NPT 6.2 h

[WORKED x2] Reduced trip speed and added asphaltic shale stabilizer
[WORKED x1] Raised MW by 0.3 ppg for borehole stability, high-vis sweeps
[FAILED x1] High-rate washing caused further wellbore enlargement
[FAILED x1] Low MW (9.2 ppg) insufficient to support Kopili shale

Barail Coal (top 3480 m) - caving: 3/14 wells, avg NPT 6.8 h

[WORKED x2] Raised MW by 0.3 ppg for borehole stability, high-vis sweeps
[WORKED x1] Reduced trip speed and added asphaltic shale stabilizer
[FAILED x2] High-rate washing caused further wellbore enlargement

Kopili Shale (top 4150 m) - caving: 4/14 wells, avg NPT 7.1 h

[WORKED x3] Raised MW by 0.3 ppg for borehole stability, high-vis sweeps
[WORKED x1] Reduced trip speed and added asphaltic shale stabilizer
[FAILED x1] High-rate washing caused further wellbore enlargement
[FAILED x1] Low MW (9.2 ppg) insufficient to support Kopili shale

Kopili Shale (top 4150 m) - kick: 3/14 wells, avg NPT 10.0 h

[WORKED x2] Shut-in, Driller's method circulation, raised MW by 0.4 ppg
[WORKED x1] Flow check every stand while drilling through gas-bearing sands
[FAILED x2] Pumping out of hole at high speed swabbed in gas

Namsang (top 450 m) - loss: 2/14 wells, avg NPT 3.0 h

[WORKED x1] Pumped 40 bbl CaCO₃ fine/medium LCM pill, reduced flow rate by 15%
[WORKED x1] Lowered ECD by reducing pump rate; bridged with graded LCM
[FAILED x1] Coarse nut-plug LCM plugged MWD tool, forced trip
[FAILED x1] Continued drilling blind without LCM led to total loss